EXPLAINABLE AI

Overall Explanation:

This project is a hands-on Explainable AI (XAI) project built around one central question:
Can we look inside a deep learning model and understand what it has actually learned?
The model used is ResNet18 — a well-known convolutional neural network with 18 layers, pretrained on ImageNet (a dataset of 1.2 million images across 1000 classes). It already knows how to classify bananas with high confidence. The project does not train any model from scratch. Instead, it uses the already-trained weights and runs gradients backward through the network to extract visual explanations.

The Core Idea: Gradients as Explanations

Neural networks work by taking an input, passing it forward through layers of learned weights, and producing a prediction. During training, backpropagation computes how much each weight contributed to the error and adjusts it. This project repurposes that same backward-pass mechanism — not to update weights, but to answer interpretability questions.
Specifically, if you ask "how much does each input pixel affect the banana class score?" — that is exactly what a gradient tells you. The gradient of the output score with respect to the input image is a map of pixel-level importance. This is the mathematical foundation of both tasks.

Task 1 — Vanilla Gradient Saliency Map

The input is a real photograph of a banana. The model predicts it as a banana with high confidence. Then, keeping the model frozen, the code computes:
∂(banana score) / ∂(input pixel)

for every single pixel in the 224×224 image. This derivative tells you — if this pixel were slightly brighter or darker, how much would the banana score change? Pixels with large gradients are the ones the model "cares about" most when making its decision.
The result is a saliency map — a heatmap the same size as the input image where bright regions indicate pixels that most influenced the prediction. What you observe is that the model pays most attention to edges and contours of the banana rather than filling in the entire object region. This is a known behavior of vanilla gradients — they capture boundary contrast well but are noisy and unstable in uniform regions due to gradient saturation.

Task 2 — Activation Maximization via Gradient Ascent

This task flips the question entirely. Instead of asking what the model sees in a real image, it asks: what image would make the model most certain it is seeing a banana?
The process starts from a blank canvas of random noise — an image the model currently sees as nothing in particular.

Then, using gradient ascent (the opposite of the gradient descent used in training), the pixel values themselves are iteratively updated to increase the banana class score:
image = image + learning_rate × ∂(banana score) / ∂(image)
The model weights are completely frozen. Only the pixel values change. After 300 iterations, the image has morphed into whatever pattern maximally activates the banana neuron inside ResNet18.

What emerges is not a clean banana shape. Instead, it is a swirling pattern of yellow-green curved textures — revealing that ResNet18 has learned to associate bananas with color and texture statistics rather than geometric shape. This is a well-documented phenomenon called CNN texture bias, and activation maximization is one of the clearest ways to demonstrate it.

What the Two Tasks Together Reveal

Task 1 and Task 2 are complementary probes of the same model from opposite directions:
Task 1 starts from a real image and asks what the model noticed. Task 2 starts from noise and asks what the model wishes it could see. The fact that Task 1 highlights edges while Task 2 produces textures tells a coherent story — ResNet18 uses edge boundaries as discriminative features in real images, but its internal representation of "banana" is a yellow curved texture pattern, not a shape. The model has learned a texture shortcut that works well on ImageNet but is quite different from how humans conceptualize objects.
This gap between what the model uses and what humans expect is precisely the problem that Explainable AI exists to surface — and this project demonstrates it clearly using nothing more than two backward passes through a standard pretrained network.

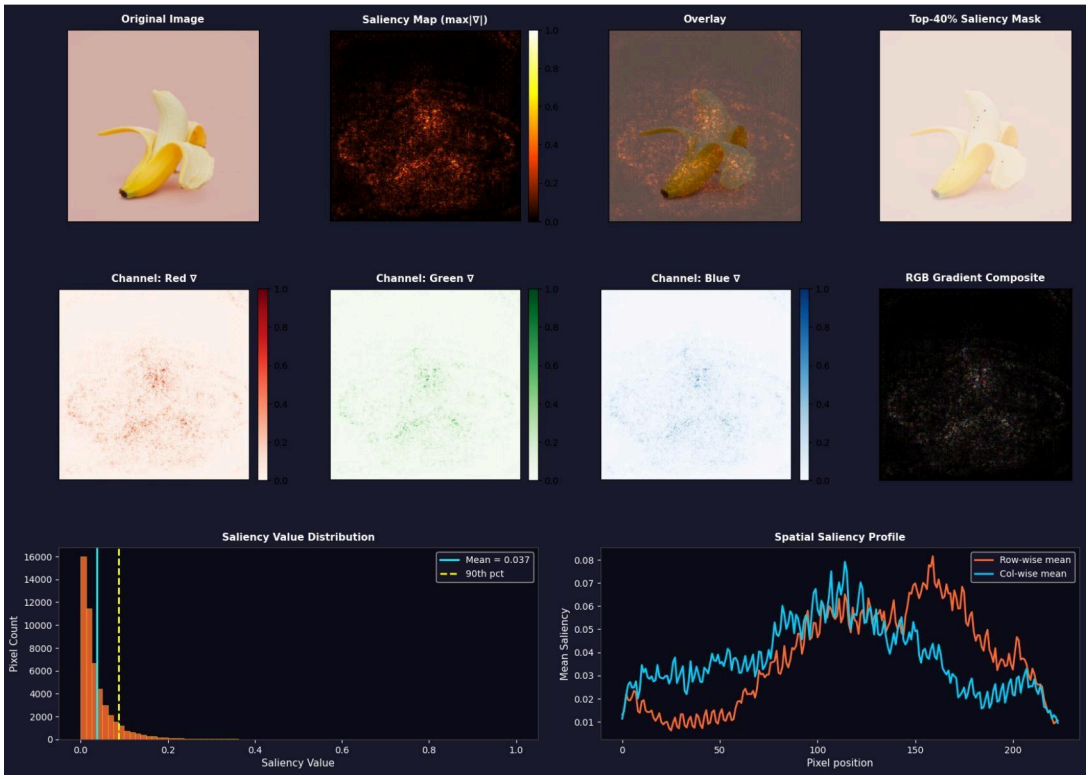
Code Cell 6:

```
Top-5 Predictions (pretrained ResNet18)
──────────────────────────────────────────────────────────
  1. [ 954] banana                          45.88%  ▇▇▇▇▇▇▇▇▇▇▇▇▇▇▇▇▇▇▇▇
  2. [ 986] class_986                       14.17%  ▇▇▇▇▇
  3. [ 951] lemon                            7.85%  ▇▇▇
  4. [ 584] class_584                        3.14%  ▇
  5. [ 112] class_112                        2.39%

  Banana probability : 45.88%
```

Prints the model's top-5 class predictions with confidence percentages displayed as ASCII bars. This is a sanity check — it confirms ResNet18 correctly identifies the image as a banana before any gradient analysis begins. If banana doesn't rank highly here, the saliency maps that follow would be meaningless.

Code Cell 8 (Task 1):



Top row:

Original Image — the banana photo resized to 224×224, shown for direct comparison.

Saliency Map (max|∇|) — a hot-yellow heatmap where brighter pixels mean "this pixel strongly influenced the banana prediction." You can see the glow concentrating along the banana's outline and peel edges rather than filling its interior cleanly.

Overlay — the saliency heatmap blended at 55% opacity on top of the original. This makes it immediately clear that the model focuses on the banana's body edges and the peeled-peel boundary, while the pink background stays relatively dark.

Top-40% Saliency Mask — a binary red mask showing only the most activated pixels. This isolates the highest-response region, showing it tracks the banana shape rather than being randomly scattered.

Bottom row:

Red Channel Gradient — how much the red pixels individually drove the prediction. Strong response because banana yellow has high red content.

Green Channel Gradient — similarly strong, since yellow = red + green.

Blue Channel Gradient — noticeably weaker, as bananas contain little blue.

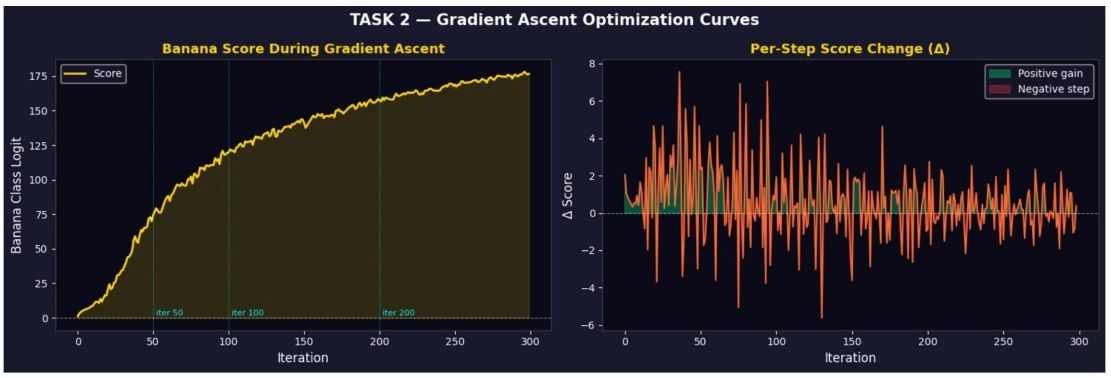
RGB Gradient Composite — all three channels stacked into one color image, showing the combined texture of gradients across the banana region.

Statistics row:

Saliency Histogram — distribution of all 224×224 saliency values. The bulk of pixels cluster near zero (background and interior banana pixels), with a sparse tail of high-saliency values at the edges. The cyan mean line and yellow 90th-percentile line mark the statistical spread.

Spatial Profile — row-wise and column-wise mean saliency plotted as line graphs. Shows where vertically and horizontally across the image the model is paying most attention — the peaks roughly correspond to the banana's position in the frame.

Code Cell 11:



Two side-by-side charts tracking the optimization progress over 300 iterations:

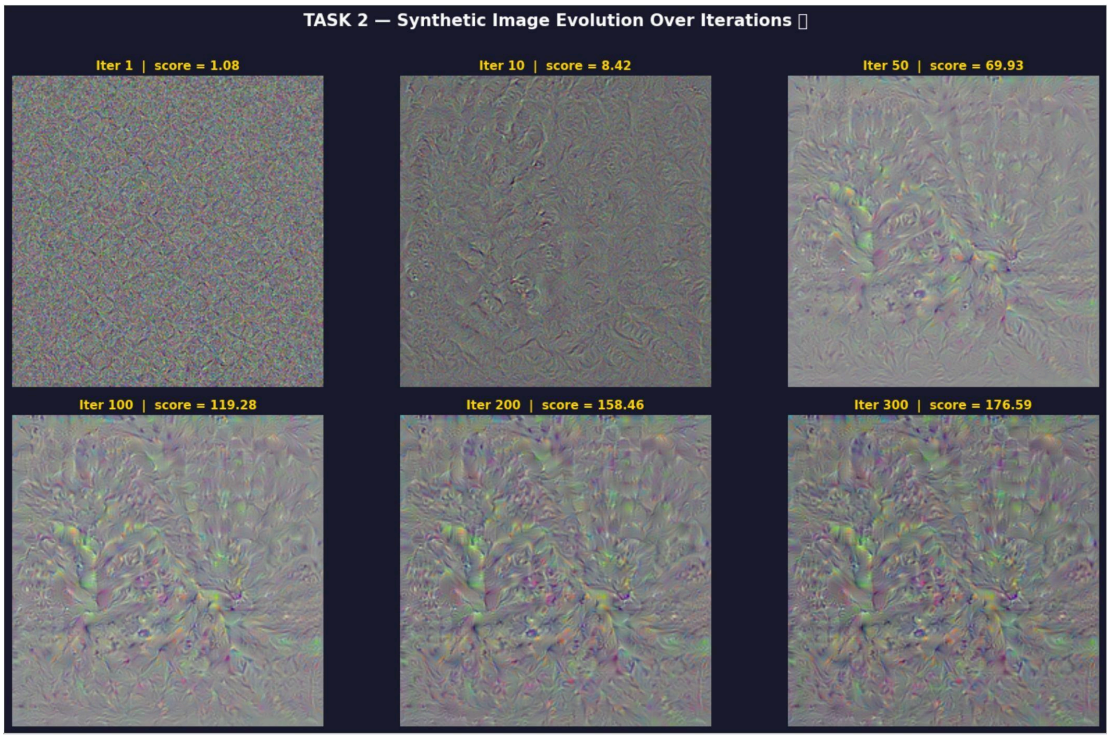
Left

Banana Score Over Time: The banana class logit starts near zero (random noise has no class signal) and rises steadily, confirming the gradient ascent is successfully pushing the

image toward what the model considers a banana. The curve rises steeply early then flattens as the optimization converges. Vertical cyan lines mark iterations 50, 100, and 200 as checkpoints.

Right
Per-Step Score Change (Δ): Shows how much the score improves at each individual step. Green fills indicate steps where the score went up; red fills mark occasional small drops. Large positive deltas early in training shrink toward zero by iteration 300, confirming convergence. Together these two charts prove the optimization is both working and stable.

Code Cell 12:

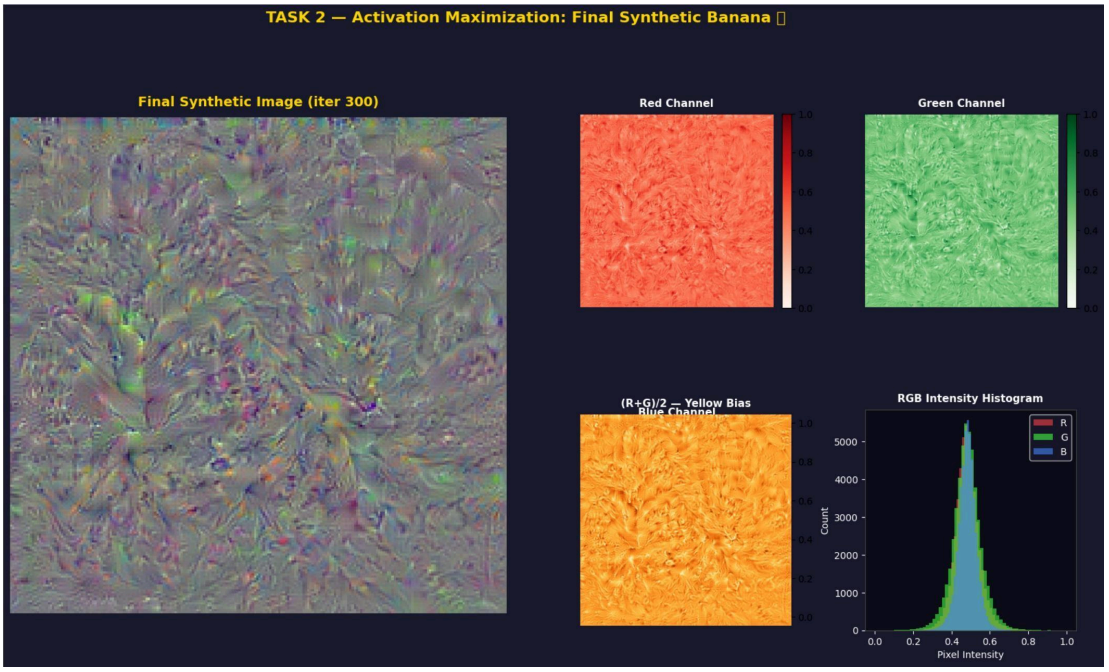


A 2×3 grid of snapshots saved at iterations 1, 10, 50, 100, 200, and 300, with the banana score printed on each panel. This is the most visually intuitive output of the project —

you can literally watch the model's banana concept emerge from noise:

- Iter 1: Pure random static — grey speckled noise with no structure whatsoever.
- Iter 10: Faint yellow-green smearing begins — the model starts biasing toward warm tones.
- Iter 50: Blobs and patches of yellow-green appear — crude but recognizable as something warm-colored.
- Iter 100: Curved, swirling textures form — the model is now strongly imprinting its internal banana representation.
- Iter 200–300: A stable psychedelic pattern of yellow-green curved swirls — this is what ResNet18's banana neuron "wants to see." No clean banana shape, just texture.

Code Cell 13:



A detailed breakdown of the iteration-300 synthetic image:

- Full synthetic image (large, left): the complete final pattern, showing yellow-green swirling textures with no recognizable banana shape: confirming CNN texture bias.
- Red channel: bright throughout, showing strong warm-tone activation.
- Green channel: also strong, together with red producing the dominant yellow hue.
- Blue channel: visibly dimmer, confirming the model learned bananas are not blue.
- RGB Intensity Histogram: quantifies the above: R and G peaks sit higher than B, making the color bias mathematically explicit rather than just visually apparent.

Task-1:

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║         TASK 1 — Observations & Analysis                          ╕
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║                                                                   ╕
║  1. EDGE & CONTOUR DOMINANCE                                      ╕
║     The saliency map lights up most strongly along the edges      ╕
║     and outlines of the banana. This shows that the model uses    ╕
║     boundary contrast as a key classification cue.                ╕
║                                                                   ╕
║  2. NOISY INTERIOR                                                ╕
║     The interior of the banana shows scattered, noisy activations╕
║     rather than a clean filled region. This is a known artifact   ╕
║     of vanilla gradients — they saturate and become unstable.     ╕
║                                                                   ╕
║  3. BACKGROUND ACTIVATIONS                                        ╕
║     Some background pixels also show non-zero saliency, meaning   ╕
║     the model is partially influenced by surrounding context.     ╕
║                                                                   ╕
║  4. RGB CHANNEL DIFFERENCES                                       ╕
║     The Red and Green channels carry more gradient signal than    ╕
║     Blue — consistent with a banana's yellow color (R+G = yellow)╕
║                                                                   ╕
║  5. VERDICT                                                       ╕
║     Vanilla Gradient Saliency highlights EDGES more than filled   ╕
║     OBJECT REGIONS. It is a fast but noisy interpretability       ╕
║     method. SmoothGrad / Integrated Gradients give cleaner maps.  ╕
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Task-2:

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║         TASK 2 — Activation Maximization Observations             ╒
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║                                                                   ╒
║  1. YELLOW / CURVED TEXTURES                                      ╒
║     The synthetic image develops yellowish, curved, blob-like     ╒
║     patterns — exactly the colour signature of a banana.          ╒
║     The R and G channels dominate over B (R+G = yellow).          ╒
║                                                                   ╒
║  2. TEXTURE BIAS OF CNN                                           ╒
║     The pattern is repetitive and texture-rich rather than a      ╒
║     clean banana shape. This confirms that ResNet18 (like most    ╒
║     CNNs) is texture-biased, not shape-biased.                    ╒
║                                                                   ╒
║  3. MONOTONIC SCORE INCREASE                                      ╒
║     The banana logit rises steadily from ~0 (random noise) to     ╒
║     a high positive value, confirming the ascent is working.      ╒
║                                                                   ╒
║  4. NOISE WITHOUT REGULARIZATION                                  ╒
║     Without total-variation or L2 regularization the image stays  ╒
║     noisy and high-frequency. Adding regularization would yield   ╒
║     smoother, more interpretable patterns.                        ╒
║                                                                   ╒
║  5. INTERPRETATION                                                ╒
║     This is the model's internal "prototype" for banana —         ╒
║     a form of model introspection / deep dream visualization.     ╒
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